Real-Time Environmental & Crop Health Monitoring System

"AgriSense" - Software-Based Precision A

PROBLEM STATEMENT

Traditional crop monitoring depends on manual field observation, which is slow, inaccurate and cannot detect environmental stress in time. Farmers have no affordable way to track environmental and crop health parameters in real time, which leads to delayed decisions, crop losses and wastage of water and resources. There is a need for a low-cost, software-based system that continuously monitors environmental parameters and crop health in real time and provides timely alerts and recommendations.

Key Problems Identified:

1. **Manual monitoring is outdated**
   Farmers still check crop and field conditions by physical observation, which is slow, irregular and cannot provide continuous information about field conditions.

2. **No real-time environmental data**
   Critical parameters such as air temperature, relative humidity, rainfall, wind speed, soil moisture and soil temperature are not tracked continuously, so stress conditions are noticed too late.

3. **Late detection of crop stress**
   Heat stress, cold stress and drought are usually identified only after visible damage appears on the crop, when recovery is already difficult or impossible.

4. **Disease-favourable conditions go unnoticed**
   Fungal diseases (blight, mildew, rust) develop under specific conditions such as high humidity combined with warm temperature. Without monitoring, preventive action cannot be taken in time.

5. **Water wastage due to guesswork irrigation**
   Without soil moisture data, farmers either over-irrigate (wasting water and causing waterlogging) or under-irrigate (causing crop stress and yield loss).

6. **Existing solutions are expensive**
   Current precision-agriculture systems depend on IoT sensor kits that are costly to buy, install and maintain, and are not affordable for small and marginal farmers.

7. **Lack of an affordable software-only platform**
   There is no easy-to-use, software-only system that monitors environmental and crop health parameters in real time and provides crop-specific health assessment, alerts and recommendations - without any hardware investment.

Hence, the proposed system AgriSense addresses the above problems by providing a real-time, software-only monitoring dashboard for environmental and crop health parameters with crop-specific alerts and recommendations.